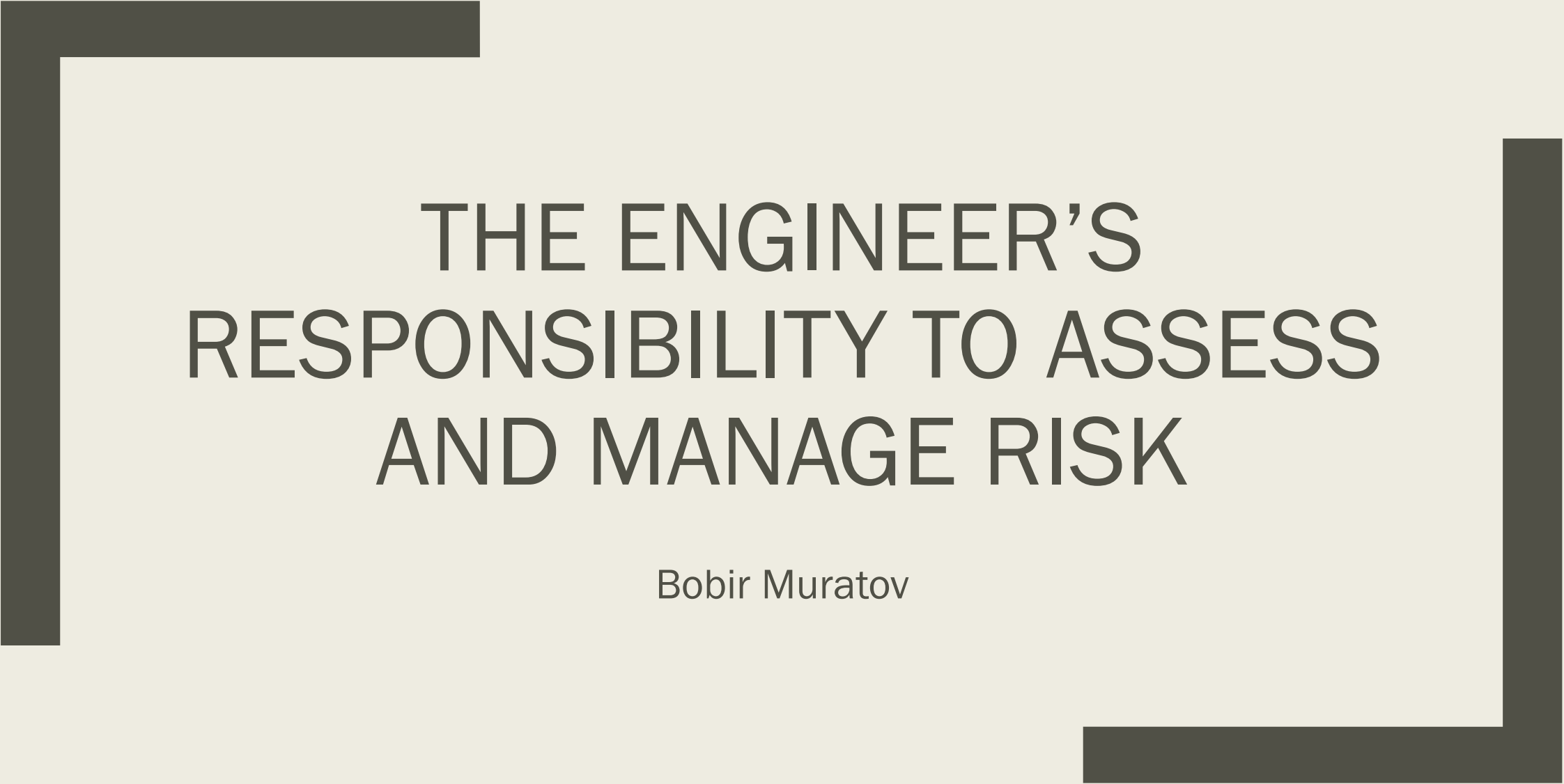


A dark gray L-shaped frame is positioned on the left and right sides of the slide, framing the central text. The left part of the frame is a vertical line extending from the top to the bottom, and the right part is a vertical line extending from the top to the bottom, with horizontal bars at the top and bottom connecting them.

THE ENGINEER'S RESPONSIBILITY TO ASSESS AND MANAGE RISK

Bobir Muratov

Case



Case



Introduction

- How these structural failures could have happened, why the building codes did not better protect the public, and how to reduce the risk of such disasters in the future
- How should engineers deal with issues of safety and risk, especially when they involve possible liability for harm?
- While engineering and construction practices change gradually over time, engineering practices also change as risks change or as our understanding of risks changes
- Safety and risk -> cost

The Engineers Approach to Risk

- An Engineering Definition of Risk:
- To assess a risk of harm, an engineer must identify it and quantify it
- Risk – the product of the probability (p_i) of a harm and the magnitude (h_i) of that harm
- The summation notation then implies the summation of all risk components with similar harms

$$Risk = \sum_{i=1}^n p_i h_i$$

The Engineers Approach to Risk

- **How Engineers Impose and Manage Risks:**
- Risk is managed in engineering design by design codes – rules proven to produce designs consistent with accepted engineering practice and which do not introduce unacceptable risks
- These design rules usually focus on proportioning the system so that the capacity (strength) of the design exceeds the demands (loads) by a specified margin
- Risk is also managed in the operation of engineering systems by development of and adherence to proven operational and maintenance rules.

The Engineers Approach to Risk

- What are the Sources of Risks on Engineering Systems?

The Engineers Approach to Risk

- **Sources of Risks Managed by Engineers:**
- Environmental loadings resulting from weather events, seismic events, or even cosmic events, and human actions, both unintentional and intentional
- Human error in the design process leading to faulty design
- Engineers are also concerned with attacks on engineered facilities, including both kinetic actions by terrorists and cyberattacks
- Risk analyses should also include identification of both threats and vulnerabilities
- Risks are dynamic; actual risks can change during the lifetime of an engineered system
- Risk is generally increased by innovation in both engineering design and in operations

The Engineers Approach to Risk

- One Engineering Approach to Defining Acceptable Risk:
- Cost-benefit analysis
- The engineering solution that maximizes net benefits (benefits minus costs) consistent with economic and other constraints is typically selected
- Compare the cost associated with the risk to the cost of preventing or drastically reducing it
- Limitations?

The Engineers Approach to Risk

- Expanding the Engineering Account of Risk: The Capabilities Approach to Identifying Harm and Benefit
- What is capabilities approach?

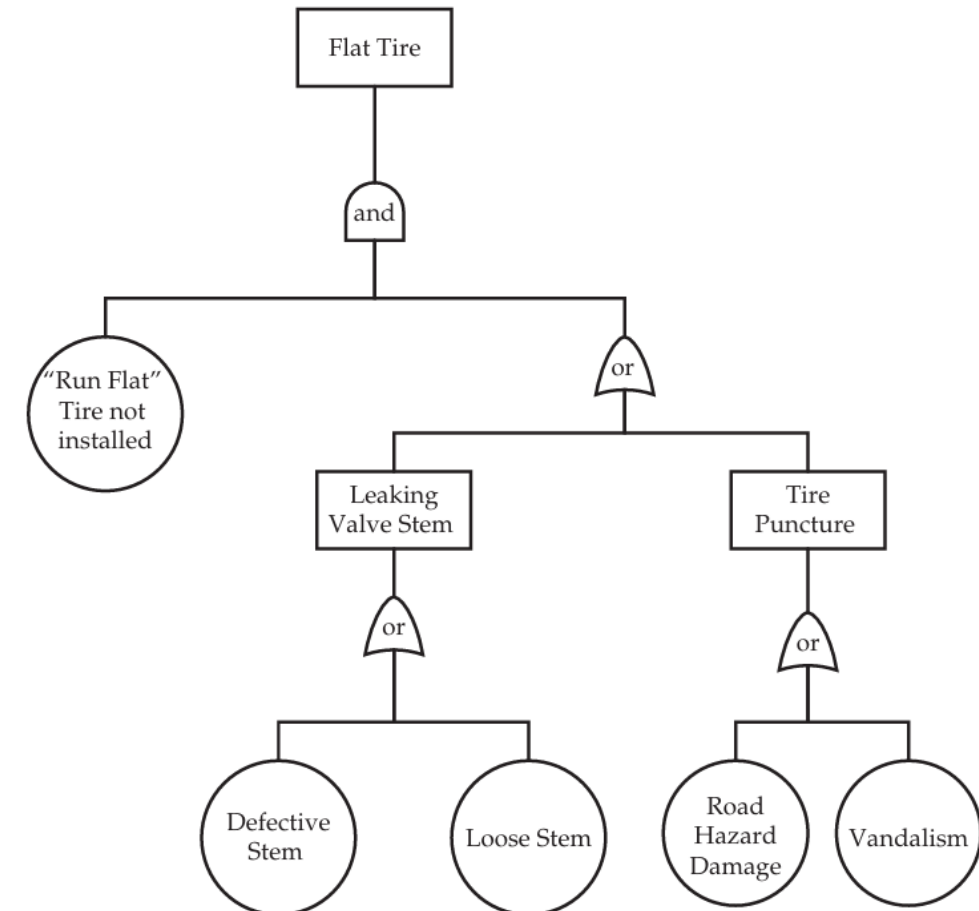
The Engineers Approach to Risk

- **Expanding the Engineering Account of Risk: The Capabilities Approach to Identifying Harm and Benefit:**
- Focuses on the effect of disasters on overall human well-being
- Well-being is defined in terms of individual capabilities or “the ability of people to lead the kind of life they have reason to value.”
- Capabilities are distinct from utilities, which refer to the mental satisfaction, pleasure, or happiness of a particular individual
- From the capabilities standpoint, a risk is the probability that individuals capabilities might be reduced due to some hazard

Difficulties in Determining the Causes and Likelihood of Harm: the Critical Attitude

- Limitations in Identifying Failure Modes
- Fault tree analysis (FTA), a formal backward looking deductive analysis, to determine the immediate and basic causes of some undesirable event
- Limitations?

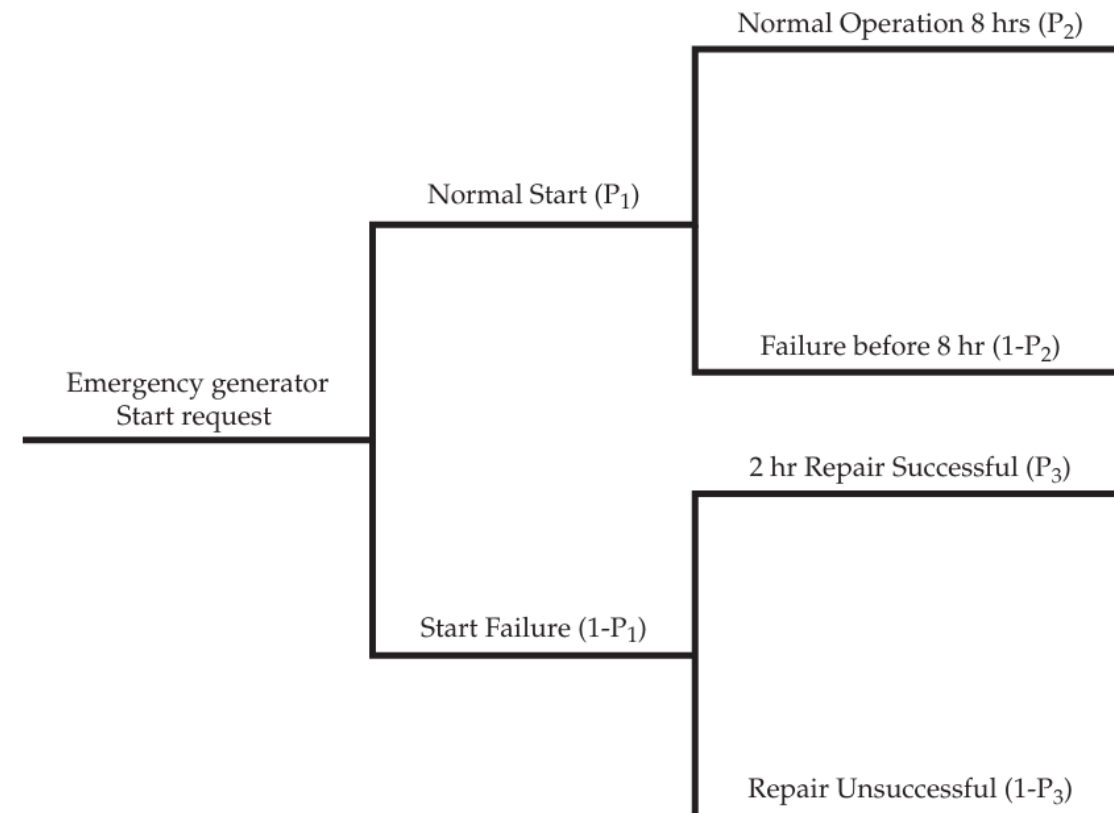
BOX 6.2 Fault Tree Example



Difficulties in Determining the Causes and Likelihood of Harm: the Critical Attitude

- Limitations in Identifying Failure Modes
- Event tree analysis (ETA) – we reason forward from a hypothetical initiating event to determine what consequences that initiating event might have and then estimate the probabilities of these consequences
- Limitations?

BOX 6.3 Event Tree Analysis Example



Difficulties in Determining the Causes and Likelihood of Harm: the Critical Attitude

- **Limitations due to Tight Coupling and Complex Interactions:**
- Two characteristics of high-risk technologies that make them especially susceptible to accidents and allow us to speak of normal accidents:
- Tight coupling - processes are tightly coupled, one process is known to affect another and will usually do so within a short time. Little time to correct a failure and little likelihood of confining a failure to one part of the system
- Complex interaction - the inability to predict consequences is crucial. Processes can be complexly interactive in that the parts of the system can interact in unanticipated ways
- These factors make accidents not only more likely but also more difficult to predict and control. This, in turn, makes risk more difficult to estimate

Difficulties in Determining the Causes and Likelihood of Harm: the Critical Attitude

- **Normalizing Deviance and Self-Deception:**
- Normalization of deviance - factor that increases risk and also decreases our ability to anticipate harm is increasing the allowable deviations from proper standards of safety and acceptable risk
- Some design predictions are not fulfilled – anomalies
- Some engineers simply accept the anomaly or even increase the boundaries of acceptable risk
- Challenger disaster

The Publics Approach to Risk

- **Expert and Layperson: Differences in Factual Beliefs:**
- Why does an engineer need to understand these differences?
- The engineer, when quantifying risks and benefits, must remember to think about the publics understanding and acceptance of the risks that the engineers work will impose and know that it may be very different from the way engineers assess risks.
- What are the grounds for these profound differences in outlook on risk?
- Engineers and risk experts believe that the public is sometimes mistaken in estimating the probability of death and injury from various activities or technologies

The Publics Approach to Risk

- **Risky Situations and Acceptable Risk:**
- Public combines concepts of risk and acceptable risk, engineers separate them
- “Risky” – the function of a warning sign, a signal that special care should be taken in a certain area
- New and unfamiliar situations, a questionable source
- Voluntarily assumed risks are more acceptable than risks not voluntarily assumed

The Publics Approach to Risk

- **Free and Informed Consent:**
- To give free and informed consent to the risks imposed by technology:
 - a person must not be coerced
 - a person must have the relevant information
 - a person must be rational and competent enough to evaluate the information
- But there are limitations?

The Publics Approach to Risk

- **Free and Informed Consent:**
- But there are limitations:
 - It is difficult to know when consent is free
 - People are often not adequately informed of dangers or do not evaluate them correctly
 - It is often not possible to obtain meaningful informed consent from individuals who are subject to risks from technology
- We could compensate individuals after the fact for actual harms done to them through technology
- Problems?

The Publics Approach to Risk

- **Equity and Justice:**
- The ethics of respect for persons places great emphasis on respecting the moral agency of individuals, regardless of the cost to the larger society
- Mr. Talbert case
- Utilitarian perspective
- It is not possible to distribute all risks and benefits equally, but...
- An acceptable risk is one in which (1) risk is assumed by free and informed consent, or properly compensated, and in which (2) risk is justly distributed, or properly compensated.

Communicating Risk And Public Policy

■ Communicating Risk to the Public:

BOX 6.5 Communicating with the Public About Risks

- Use familiar terminology—"probability of harm" may be more clearly understood than "risk."
- Qualitatively compare "new" risks by comparison to "familiar" risks. "The probability of flooding related to the new development should not be greater than the present probability of flooding."
- Acknowledge uncertainty in risk assessments.
- Recognize that costs versus benefits are not the only factor in determining acceptability of risks.
- Be "objective and truthful" in all public statements.

Becoming A Responsible Engineer Regarding Risk

- Principle of acceptable risk:
- People should be protected from the harmful effects of technology, especially when the harms are not consented to or when they are unjustly distributed, except that this protection must sometimes be balanced against (1) the need to preserve great and irreplaceable benefits, and (2) the limitation on our ability to obtain informed consent.